

Problem 5

计算下列积分:

(1) $\int_0^{2\pi} \cos^{2n} \theta d\theta$, n 为正整数;

解 令 $z = e^{i\theta}$, 则 $\cos \theta = \frac{z+z^{-1}}{2}$, $d\theta = \frac{dz}{iz}$, 代入得

$$\begin{aligned} I &= \oint_{|z|=1} \left(\frac{z+z^{-1}}{2} \right)^{2n} \frac{dz}{iz} = \frac{1}{2^{2n}i} \oint_{|z|=1} \frac{(z^2+1)^{2n}}{z^{2n+1}} dz \\ &= \frac{1}{2^{2n}i} \cdot 2\pi i \cdot \frac{1}{(2n)!} \lim_{z \rightarrow 0} \frac{d^{2n}}{dz^{2n}} [(z^2+1)^{2n}] \\ &= \frac{\pi}{2^{2n-1}(2n)!} \lim_{z \rightarrow 0} \frac{d^{2n}}{dz^{2n}} \left[\sum_{k=0}^{2n} C_{2n}^k (z^2)^{2n-k} \right] \\ &= \frac{\pi}{2^{2n-1}(2n)!} \cdot (2n)! C_{2n}^n = \frac{\pi}{2^{2n-1}} C_{2n}^n. \end{aligned}$$

若 $R(\cos \theta, \sin \theta)$ 为 θ 的偶函数, 则 $\int_0^\pi R(\cos \theta, \sin \theta) d\theta$ 型积分亦可由上述方法求解, 因为 $\int_0^\pi R(\cos \theta, \sin \theta) d\theta = \frac{1}{2} \int_{-\pi}^\pi R(\cos \theta, \sin \theta) d\theta$.

(2) $\int_0^{2\pi} \frac{dx}{(a+b \cos x)^2}$, $a > b > 0$;

令 $f(z) = \frac{1}{z \left(a + b \frac{z^2+1}{2z} \right)^2} = \frac{4}{b^2} \frac{z}{(z-z_1)^2(z-z_2)^2}$, 其中 $z_1 = -\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1}$,

$z_2 = -\frac{a}{b} + \sqrt{\left(\frac{a}{b}\right)^2 - 1}$. $-\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1} < -\frac{a}{b} < -1$, 所以 $|z_1| > 1$, 在单位圆外,

$|z_2| = \left| -\frac{a}{b} + \sqrt{\left(\frac{a}{b}\right)^2 - 1} \right| = 1 / \left| -\frac{a}{b} - \sqrt{\left(\frac{a}{b}\right)^2 - 1} \right| < 1$, 在单位圆内.

$\text{res } f(z_2) = \lim_{z \rightarrow z_2} \frac{d}{dz} [(z-z_2)^2 f(z)] = \frac{a}{(a^2-b^2)^{3/2}},$

所以原积分 $= \frac{2\pi a}{(a^2-b^2)^{3/2}}.$

(3) $\int_0^\pi \frac{d\theta}{1+\sin^2 \theta};$

$$\text{原积分} = \int_0^\pi \frac{2d\theta}{3 - \cos 2\theta} = \int_0^{2\pi} \frac{d\varphi}{3 - \cos \varphi}$$

(3)

Proof. We note $\sin^2 \theta = \frac{1}{2}[1 - \cos(2\theta)]$. Using the substitution rule $\theta = \frac{1}{2}\alpha$, we have

$$\int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} = \int_0^{2\pi} \frac{d\alpha}{3 - \cos \alpha} = 2i \int_{|z|=1} \frac{dz}{z^2 - 6z + 1}.$$

The equation $z^2 - 6z + 1 = 0$ has two roots: $z_1 = 3 + 2\sqrt{2}$ and $z_2 = 3 - 2\sqrt{2}$. It's clear that $|z_1| > 1$ and $|z_2| < 1$. By Residue Theorem, we have

$$\int_0^\pi \frac{d\theta}{1 + \sin^2 \theta} = 2i \cdot \frac{2\pi i}{2\pi i} \int_{|z|=1} \frac{dz}{(z - z_1)(z - z_2)} = -4\pi \cdot \frac{1}{z_2 - z_1} = \frac{\pi}{\sqrt{2}}.$$

□

$$(4) \int_0^\pi \frac{d\theta}{(1 + \sin^2 \theta)^2}.$$

$$\text{原积分} = \int_0^\pi \frac{4d\theta}{(3 - \cos 2\theta)^2} = \int_0^{2\pi} \frac{2d\varphi}{(3 - \cos \varphi)^2}$$

$$\text{令 } f(z) = \frac{1}{z} \frac{2}{\left(3 - \frac{z^2 + 1}{2z}\right)^2} = \frac{8z}{(z - 3 - 2\sqrt{2})^2 (z - 3 + 2\sqrt{2})^2}, \text{ 则}$$

$$\text{res } f(3 - 2\sqrt{2}) = \lim_{z \rightarrow 3 - 2\sqrt{2}} \frac{d}{dz} \frac{8z}{(z - 3 - 2\sqrt{2})^2} = \frac{3}{8\sqrt{2}}, \text{ 原积分} = \frac{3}{4\sqrt{2}} \pi$$

Problem 6

计算下列积分:

$$(1) \int_{-\infty}^{\infty} \frac{x^2}{1 + x^4} dx;$$

$$\text{令 } f(z) = \frac{z^2}{1 + z^4} = \frac{z^2}{(z - e^{i\pi/4})(z - e^{-i\pi/4})(z - e^{i3\pi/4})(z - e^{-i3\pi/4})}$$

$$\operatorname{res} f(e^{i\pi/4}) = \frac{1}{4\sqrt{2}}(1-i), \quad \operatorname{res} f(e^{i3\pi/4}) = -\frac{1}{4\sqrt{2}}(1+i)。$$

$$\text{所以 v.p.} \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx = 2\pi i \left[\operatorname{res} f(e^{i\pi/4}) + \operatorname{res} f(e^{i3\pi/4}) \right] = \frac{1}{\sqrt{2}} \pi。$$

$$\begin{aligned} \text{被积函数是偶函数, 所以 } \int_0^{\infty} f(x) dx &= \frac{1}{2} \lim_{b \rightarrow \infty} 2 \int_0^b f(x) dx = \frac{1}{2} \lim_{b \rightarrow \infty} \left[\int_{-b}^0 f(x) dx + \int_0^b f(x) dx \right] \\ &= \frac{1}{2} \text{v.p.} \int_{-\infty}^{\infty} f(x) dx, \text{ 即 } \int_0^{\infty} f(x) dx \text{ 收敛, 同样的, } \int_{-\infty}^0 f(x) dx \text{ 也收敛, 所以 } \int_{-\infty}^{\infty} f(x) dx \text{ 收} \end{aligned}$$

$$\text{敛, 且等于 v.p.} \int_{-\infty}^{\infty} \frac{x^2}{1+x^4} dx。 \text{ 后面类似的讨论省略。}$$

$$(2) \int_{-\infty}^{\infty} \frac{1}{(1+x^2)^{n+1}} dx, n \text{ 为正整数;}$$

$$\text{令 } f(z) = \frac{1}{(1+z^2)^{n+1}} = \frac{1}{(z+i)^{n+1}(z-i)^{n+1}}, \text{ 则}$$

$$\operatorname{res} f(i) = \frac{1}{n!} \left[\frac{1}{(z+i)^{n+1}} \right]_{z=i}^{(n)} = \frac{1}{n!} \frac{(-1)^n (n+1)(n+2) \cdots (2n)}{(2i)^{2n+1}} = \frac{1}{2i} \frac{(2n)!}{2^{2n} (n!)^2} = \frac{1}{2i} \frac{(2n-1)!!}{(2n)!!}$$

$$\text{所以原积分} = \text{v.p.} \int_{-\infty}^{\infty} \frac{dx}{(1+x^2)^{n+1}} = 2\pi i \operatorname{res} f(i) = \frac{(2n-1)!!}{(2n)!!} \pi。$$

$$(3) \int_{-\infty}^{\infty} \frac{x^{2m}}{1+x^{2n}} dx, \quad n, m \in \mathbb{Z} \text{ 且 } n > m$$

$$\text{令 } f(z) = \frac{z^{2m}}{z^{2n}+1}, \text{ 它在上半平面的奇点是 } e^{i(k+\frac{1}{2})\frac{\pi}{n}} \quad (k=0,1,2,\dots,n-1)。$$

$$\begin{aligned} \operatorname{res} f \left[e^{i(k+\frac{1}{2})\frac{\pi}{n}} \right] &= \lim_{z \rightarrow e^{i(k+\frac{1}{2})\frac{\pi}{n}}} \frac{z^{2m}}{(z^{2n}+1)'} = \lim_{z \rightarrow e^{i(k+\frac{1}{2})\frac{\pi}{n}}} \frac{z^{2m-2n+1}}{2n} \\ &= \frac{1}{2n} e^{ik(2m-2n+1)\frac{\pi}{n}} e^{i(2m-2n+1)\frac{\pi}{2n}} \end{aligned}$$

$$\text{原积分} = \text{v.p.} \int_{-\infty}^{\infty} \frac{x^{2m}}{x^{2n}+1} dx$$

$$\begin{aligned} &= 2\pi i \sum_{k=0}^{n-1} \operatorname{res} f \left[e^{i(k+\frac{1}{2})\frac{\pi}{n}} \right] = \frac{\pi i}{n} e^{i(2m-2n+1)\frac{\pi}{2n}} \sum_{k=0}^{n-1} e^{ik(2m-2n+1)\frac{\pi}{n}} \\ &= \frac{\pi i}{n} e^{i(2m-2n+1)\frac{\pi}{2n}} \frac{e^{i(2m-2n+1)\pi} - 1}{e^{i(2m-2n+1)\frac{\pi}{n}} - 1} = -\frac{\pi i}{n} \frac{2}{2i \sin \frac{(2m-2n+1)\pi}{2n}} \\ &= \frac{\pi}{n \sin \frac{(2m+1)\pi}{2n}}。 \end{aligned}$$

Proof. We note

$$\int_{-\infty}^{\infty} \frac{x^{2m}}{1+x^{2n}} dx = 2 \int_0^{\infty} \frac{x^{2m}}{1+x^{2n}} dx = \frac{1}{n} \int_0^{\infty} \frac{y^{\frac{2m+1}{2n}}}{(1+y)y} dy$$

by the substitution rule $x^{2n} = y$. Define $p = \frac{2n}{2m+1}$, then $p > 1$. In our solution of exercise 7.8 in the text, we already showed

$$\int_0^{\infty} \frac{y^{\frac{1}{p}}}{(1+y)y} dy = \pi \csc\left(\frac{\pi}{p}\right).$$

$$\text{So } \int_{-\infty}^{\infty} \frac{x^{2m}}{1+x^{2n}} dx = \frac{\pi}{n} \csc\left(\frac{2m+1}{2n}\pi\right).$$

□

$$(4) \int_{-\infty}^{\infty} \frac{dx}{(1+x^2) \cosh \frac{\pi x}{2}}.$$

Proof. We note $\cosh(\frac{\pi}{2}z) = 0$ if and only if $z = (2n+1)i$ ($n \in \mathbb{Z}$). Define $f(z) = \frac{1}{(z^2+1) \cosh(\frac{\pi}{2}z)}$. Then for $n \neq 0, -1$, z_n is a pole of order 1 for $f(z)$ and we have (assume $\rho > 0$ is sufficiently small)

$$\text{Res}(f, z_n) = \frac{1}{2\pi i} \int_{|z-z_n|=\rho} \frac{1}{z-z_n} \cdot \frac{\frac{\pi}{2}z - \frac{\pi}{2}z_n}{\cosh(\frac{\pi}{2}z) - \cosh(\frac{\pi}{2}z_n)} \frac{dz}{\frac{\pi}{2}(z^2+1)} = \frac{1}{\sinh(\frac{\pi}{2}z_n)} \cdot \frac{1}{\frac{\pi}{2}(z_n^2+1)} = \frac{i}{2\pi} \frac{(-1)^{n+1}}{n(n+1)}.$$

To find the residue of $f(z)$ at i , define $h(z) = \frac{z-i}{2 \cosh(\frac{\pi}{2}z) - \cosh(\frac{\pi}{2}i)}$. Then $h(i) = \frac{1}{\sinh(\frac{\pi}{2}i)} = -i$. Applying l'Hospital's rule, we have

$$\begin{aligned} h'(i) &= \lim_{z \rightarrow i} \frac{\frac{\pi}{2} \cosh(\frac{\pi}{2}z) - \frac{\pi}{2}(z-i) \frac{\pi}{2} \sinh(\frac{\pi}{2}z)}{[\cosh(\frac{\pi}{2}z)]^2} = \lim_{z \rightarrow i} \frac{\frac{\pi^2}{4} \sinh(\frac{\pi}{2}z) - \frac{\pi^2}{4} [\sinh(\frac{\pi}{2}z) + (z-i) \frac{\pi}{2} \cosh(\frac{\pi}{2}z)]}{2 \cosh(\frac{\pi}{2}z) \frac{\pi}{2} \sinh(\frac{\pi}{2}z)} \\ &= \lim_{z \rightarrow i} \frac{-\frac{\pi}{2}(z-i)}{\pi \sinh(\frac{\pi}{2}z)} \cdot \frac{\pi^2}{4} = 0. \end{aligned}$$

Therefore

$$\text{Res}(f, i) = \frac{d}{dz} \left[\frac{2h(z)}{\pi(z+i)} \right] \Big|_{z=i} = \frac{2[h'(z)(z+i) - h(z)]}{\pi(z+i)^2} \Big|_{z=i} = \frac{1}{2\pi i}.$$

So

$$\sum_{n=0}^{\infty} \text{Res}(f, z_n) = \frac{1}{2\pi i} + \sum_{n=1}^{\infty} \frac{i}{2\pi} \frac{(-1)^{n+1}}{n(n+1)} = \frac{1}{2\pi i} + \frac{1}{2\pi i} (-1 + 2 \ln 2) = \frac{2 \ln 2}{2\pi i}.$$

Now we consider the path $C_N = [-N, N, N+4Ni, -N+4Ni, -N]$. Then

$$\oint_{C_N} f(z) dz = I + II + III + IV,$$

where

$$I = \int_{-N}^N f(x) dx, II = \int_0^{4N} f(N+iy) i dy, III = \int_N^{-N} f(x+4Ni) dx, IV = \int_{4N}^0 f(-N+iy) i dy.$$

We note

$$|II| \leq \int_0^{4N} \frac{dy}{(|N+iy|^2 - 1) \cdot \frac{e^{\pi N/2} - e^{-\pi N/2}}{2}} \leq \frac{2}{e^{\frac{\pi N}{2}} - e^{-\frac{\pi N}{2}}} \cdot \frac{4N}{N^2 - 1} \rightarrow 0$$

as $N \rightarrow \infty$. Similarly, we can show $|IV| \rightarrow 0$ as $N \rightarrow \infty$. Meanwhile, we have

$$|III| \leq \int_{-N}^N \frac{dx}{(|x+4Ni|^2 - 1) \frac{e^{\frac{\pi}{2} + e^{-\frac{\pi}{2}}}}{2}} \leq \frac{2N}{16N^2 - 1} \rightarrow 0$$

as $N \rightarrow \infty$. Therefore, $\int_{-\infty}^{\infty} f(x) dx = \lim_{N \rightarrow \infty} \oint_{C_N} f(z) dz$ and by Residue Theorem, we have

$$\int_{-\infty}^{\infty} f(x) dx = \lim_{N \rightarrow \infty} \oint_{C_N} f(z) dz = \lim_{N \rightarrow \infty} 2\pi i \sum_{n=0}^{2N-1} \text{Res}(f, z_n) = 2\pi i \sum_{n=0}^{\infty} \text{Res}(f, z_n) = 2\pi i \cdot \frac{2 \ln 2}{2\pi i} = 2 \ln 2.$$

Remark 9. In the proof, we used the following facts from calculus (see, for example, Shen [9], page 221):

$$\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$$

and

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)} &= \sum_{n=1}^{\infty} (-1)^n \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &= \left(\frac{1}{2} - 1 \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{4} - \frac{1}{3} \right) + \cdots \\ &= 1 - 2 \left(1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots \right) \\ &= 1 - 2 \ln 2. \end{aligned}$$

The calculus proof of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} = \ln 2$ needs a little bit trick. However, if we use theory of analytic functions, then the proof becomes straightforward. Indeed, we note in the unit disc,

$$\ln(1+z) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^n}{n}.$$

The series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^n}{n}$ clearly converges at $z=1$. So by Abel's Second Theorem (see, for example, Fang [3], page 121), we must have

$$\ln 2 = \lim_{z \in \mathbb{R}, z \rightarrow 1} \ln(1+z) = \lim_{z \in \mathbb{R}, z \rightarrow 1} \sum_{n=1}^{\infty} \frac{(-1)^{n-1} z^n}{n} = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}.$$

Problem 7

计算下列积分:

$$(1) \int_0^{\infty} \frac{\cos x}{1+x^4} dx;$$

$$(1) \text{ 令 } f(z) = \frac{e^{iz}}{1+z^4} = \frac{e^{iz}}{(z-e^{i\pi/4})(z-e^{-i\pi/4})(z-e^{i3\pi/4})(z-e^{-i3\pi/4})}.$$

$$\operatorname{res} f(e^{i\pi/4}) = \frac{e^{-1/\sqrt{2}}}{4i} e^{i\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right)}, \quad \operatorname{res} f(e^{i3\pi/4}) = \frac{e^{-1/\sqrt{2}}}{4i} e^{-i\left(\frac{1}{\sqrt{2}} - \frac{\pi}{4}\right)}.$$

$$\text{原积分} = \frac{1}{2} \text{v.p.} \int_{-\infty}^{\infty} \frac{\cos x}{1+x^4} dx$$

$$= \frac{1}{2} \operatorname{Re} \left\{ 2\pi i \left[\operatorname{res} f(e^{i\pi/4}) + \operatorname{res} f(e^{i3\pi/4}) \right] \right\} = \frac{\pi}{2} e^{-\frac{1}{\sqrt{2}}} \cos \left(\frac{1}{\sqrt{2}} - \frac{\pi}{4} \right).$$

$$(2) \int_0^{\infty} \frac{\cos x}{(1+x^2)^3} dx;$$

$$\text{令 } f(z) = \frac{e^{iz}}{(1+z^2)^3}, \text{ 奇点为 } z = \pm i, \text{ 只有 } z = i \text{ 在上半圆范围内}$$

$$\int_0^{\infty} \frac{\cos x}{(1+z^2)^3} = \frac{1}{2} \int_{-\infty}^{\infty} \frac{\cos x}{(1+z^2)^3} = \pi i \cdot \operatorname{res} f(i) = \pi i \cdot \left(-\frac{7i}{16e} \right) = \frac{7\pi}{16e}$$

$$(3) \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 - 2x + 2} dx;$$

$$\text{令 } f(z) = \frac{ze^{iz}}{z^2 - 2z + 2} = \frac{ze^{iz}}{(z-1-i)(z-1+i)}, \text{ 则 } \operatorname{res} f(1+i) = \frac{1}{\sqrt{2}ei} e^{i\left(1+\frac{\pi}{4}\right)}.$$

$$\text{v.p.} \int_{-\infty}^{\infty} \frac{x \sin x}{x^2 - 2x + 2} dx = \operatorname{Im} [2\pi i \cdot \operatorname{res} f(1+i)] = \frac{\sqrt{2}\pi}{e} \sin\left(1 + \frac{\pi}{4}\right).$$

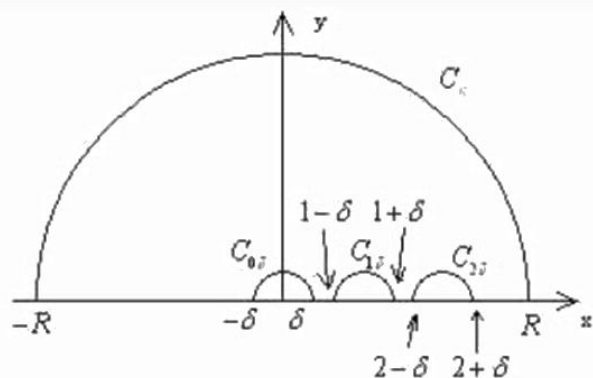
由于 $\frac{x}{x^2 - 2x + 2}$ 单调趋于 0, $\left| \int_0^b \sin x dx \right| = |1 - \cos b| \leq 2$, 所以原积分收敛 (狄里克莱判敛法), 等于其主值。

$$\begin{aligned} (4) \int_0^{\infty} \frac{\sin(a+2n)x - \sin ax}{(1+x^2) \sin x} dx, a > -1, n \in \mathbb{Z}^+ \\ &= \int_0^{\infty} \frac{\sin(a+2n)x - \sin ax}{(1+x^2) \sin x} dx = \frac{1}{2} \int_{-\infty}^{\infty} \frac{2 \cos(a+2n)x \sin nx}{(1+x^2) \sin x} dx \\ &= \int_0^{\infty} \sum_{k=0}^{n-1} \frac{\sin(a+2k+2)x - \sin(a+2k)x}{(1+x^2) \sin x} dx = \frac{n-1}{2} \int_0^{\infty} \frac{2 \cos(a+2k+1)x \sin x}{(1+x^2) \sin x} dx \\ &= \frac{n-1}{2} \int_0^{\infty} \frac{2 \cos(a+2k+1)x}{1+x^2} dx \\ &= \int_C \frac{e^{i(a+2k+1)z}}{1+z^2} dz = \int_{-R}^R \frac{e^{i(a+2k+1)z}}{1+z^2} dz + \int_{C_R} \frac{e^{i(a+2k+1)z}}{1+z^2} dz \end{aligned}$$

$$\begin{aligned} \int_{C_R} \frac{e^{i(a+2k+1)z}}{1+z^2} dz &= 0 \text{ since } \lim_{z \rightarrow \infty} \frac{z}{1+z^2} = 0 \\ \operatorname{res} f(i) &= \lim_{z \rightarrow i} \frac{e^{i(a+2k+1)z}}{z+i} = \frac{e^{i(a+2k+1)i}}{2i} \\ \text{原式} &= \sum_{k=0}^{n-1} \pi e^{-(a+2k+1)} = \pi e^{-a-1} \sum_{k=0}^{n-1} \left(\frac{1}{e^2}\right)^k = \pi e^{-a-1} \frac{1 - \left(\frac{1}{e^2}\right)^n}{1 - \frac{1}{e^2}} = \pi e^{-a-1} \frac{1 - e^{-2n}}{1 - e^{-2}} \end{aligned}$$

Problem 8

$$(1) \text{ v.p. } \int_{-\infty}^{\infty} \frac{dx}{x(x-1)(x-2)};$$



$$\text{令 } f(z) = \frac{1}{z(z-1)(z-2)}, \text{ 则 } \lim_{\delta \rightarrow 0} \int_{C_{0\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 0} \frac{1}{(z-1)(z-2)} = -\frac{\pi}{2} i,$$

$$\lim_{\delta \rightarrow 0} \int_{C_{1\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 1} \frac{1}{z(z-2)} = \pi i,$$

$$\lim_{\delta \rightarrow 0} \int_{C_{2\delta}} f(z) dz = -\pi i \lim_{z \rightarrow 2} \frac{1}{z(z-1)} = -\frac{\pi}{2} i$$

$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = \pi i \lim_{z \rightarrow \infty} \frac{1}{(z-1)(z-2)} = 0.$$

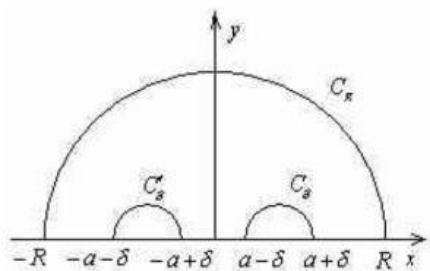
$$\begin{aligned} & \int_{-R}^{-\delta} f(x) dx + \int_{C_{0\delta}} f(z) dz + \int_{\delta}^{1-\delta} f(x) dx + \int_{C_{1\delta}} f(z) dz + \int_{1+\delta}^{2-\delta} f(x) dx + \int_{C_{2\delta}} f(z) dz \\ & + \int_{2+\delta}^R f(x) dx + \int_{C_R} f(z) dz = 0 \end{aligned}$$

$$\text{令上式 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得到 v.p.} \int_{-\infty}^{\infty} \frac{dx}{x(x-1)(x-2)} = 0.$$

$$(2) \int_0^{\infty} \frac{\sin(x+a) \sin(x-a)}{x^2 - a^2} dx, a > 0;$$

$$\text{原积分} = \frac{1}{2} \int_0^{\infty} \frac{\cos 2a - \cos 2x}{x^2 - a^2} dx. \quad \text{令 } f(z) = \frac{\cos 2a - e^{2iz}}{z^2 - a^2},$$

取如下积分路径:



$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z) dz = -\pi i \lim_{z \rightarrow a} (z-a) f(z) = \frac{\pi(\cos 2a - e^{2ia})}{2ai},$$

$$\lim_{\delta \rightarrow 0} \int_{C'_\delta} f(z) dz = -\pi i \lim_{z \rightarrow -a} (z+a) f(z) = -\frac{\pi(\cos 2a - e^{-2ia})}{2ai}.$$

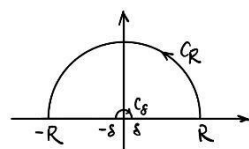
$$\lim_{R \rightarrow \infty} \int_{C_R} f(z) dz = 0.$$

$$\int_{-R}^{-a-\delta} f(x) dx + \int_{C_\delta} f(z) dz + \int_{-a+\delta}^{a-\delta} f(x) dx + \int_{C'_\delta} f(z) dz + \int_{a+\delta}^R f(x) dx + \int_{C_R} f(z) dz = 0$$

$$\text{令 } \delta \rightarrow 0, R \rightarrow \infty \text{ 得 v.p.} \int_{-\infty}^{\infty} \frac{\cos 2a - e^{2ix}}{x^2 - a^2} dx = \frac{\pi}{a} (\sin 2a - i \cos 2a).$$

$$\text{所以原积分} = \frac{1}{4} \operatorname{Re} \left(\text{v.p.} \int_{-\infty}^{\infty} \frac{\cos 2a - e^{2ix}}{x^2 - a^2} dx \right) = \frac{\pi}{4a} \sin 2a.$$

$$(3) \int_0^{\infty} \frac{x - \sin x}{x^3(1+x^2)} dx;$$



$$\int_0^{+\infty} \frac{x-\sin x}{x^3(1+x^2)} dx = I_m[\int_{-\infty}^{+\infty} \frac{ix-e^{ix}+1}{x^3(1+x^2)} dx]$$

$$\oint \frac{iz-e^{iz}+1}{z^3(1+z^2)} dz = \int_{CR} \frac{iz-e^{iz}+1}{z^3(1+z^2)} dz + \int_{-\infty}^{+\infty} f(x) dx - \int_{C_{\delta,0}} \frac{iz-e^{iz}+1}{z^3(1+z^2)} dz$$

$$\text{由Jordan定理, } \int_{CR} \frac{iz-e^{iz}+1}{z^3(1+z^2)} dz = 0$$

$$\int_{-\infty}^{+\infty} f(x) dx = \frac{i * i - e^{-1} + 1}{i^3(2i)} * 2\pi i = -\pi e^{-1}$$

$$z * f(z) = \lim_{z \rightarrow 0} \frac{iz - e^{iz} + 1}{z^2(1 + z^2)} = \lim_{z \rightarrow 0} \frac{-\frac{z^2}{2}(1 + iz + \cdots)}{z^2(1 + z^2)} = -\frac{1}{2}$$

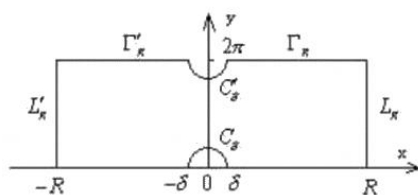
$$\int_{C_{\delta,0}} \frac{iz-e^{iz}+1}{z^3(1+z^2)} dz = \frac{1}{2} \pi i$$

$$\oint \frac{iz-e^{iz}+1}{z^3(1+z^2)} dz = \frac{1}{2} \pi i - \pi e^{-1}$$

$$\int_0^{+\infty} \frac{x-\sin x}{x^3(1+x^2)} dx = I_m[\int_{-\infty}^{+\infty} \frac{ix-e^{ix}+1}{x^3(1+x^2)} dx] = \frac{1}{2} \pi e^{-1} - \frac{1}{4} \pi$$

$$(4) \int_{-\infty}^{\infty} \frac{e^{px} - e^{qx}}{1 - e^x} dx, 0 < p < 1, 0 < q < 1.$$

令 $f(z) = \frac{e^{pz}}{1-e^z}$, $2k\pi i$ ($k=0, \pm 1, \pm 2, \dots$) 是其一阶极点。取如下积分路径:



$$\int_{-R}^{-\delta} f(x)dx + \int_{C_\delta} f(z)dz + \int_{\delta}^R f(x)dx + \int_{L_R} f(z)dz + \int_{\Gamma_R} f(z)dz + \int_{C'_\delta} f(z)dz \\ + \int_{\Gamma'_R} f(z)dz + \int_{L'_R} f(z)dz = 0. \quad (*)$$

$$\int_{\Gamma_R} f(z)dz = \int_R^\delta \frac{e^{p(x+2\pi i)}}{1-e^{(x+2\pi i)}}dx = -e^{2p\pi i} \int_\delta^R f(x)dx, \quad \int_{\Gamma'_R} f(z)dz = -e^{2p\pi i} \int_{-R}^{-\delta} f(x)dx,$$

代入(*)式得

$$(1-e^{2p\pi i}) \left[\int_{-R}^{-\delta} f(x)dx + \int_\delta^R f(x)dx \right] + \int_{C_\delta} f(z)dz + \int_{L_R} f(z)dz + \int_{C'_\delta} f(z)dz \\ + \int_{L'_R} f(z)dz = 0. \quad (**)$$

$$\left| \int_{L_R} f(z)dz \right| \leq \int_0^{2\pi} \left| \frac{e^{p(R+iy)}}{1-e^{(R+iy)}} \right| dy = \int_0^{2\pi} \frac{e^{pR}}{\sqrt{(1-e^R \cos y)^2 + e^{2R} \sin^2 y}} dy \\ = e^{pR} \int_0^{2\pi} \frac{1}{\sqrt{1-2e^R \cos y + e^{2R}}} dy \leq e^{pR} \int_0^{2\pi} \frac{1}{e^R - 1} dy = 2\pi \frac{e^{-(1-p)R}}{1-e^{-R}},$$

所以 $\lim_{R \rightarrow \infty} \int_{\Gamma_R} f(z)dz = 0$, 同样的, $\lim_{R \rightarrow \infty} \int_{\Gamma'_R} f(z)dz = 0$.

$$\lim_{\delta \rightarrow 0} \int_{C_\delta} f(z)dz = -\pi i \lim_{z \rightarrow 0} \frac{ze^{pz}}{1-e^z} = \pi i,$$

$$\lim_{\delta \rightarrow 0} \int_{C'_\delta} f(z)dz = -\pi i \lim_{z \rightarrow 2\pi i} \frac{(z-2\pi i)e^{pz}}{1-e^z} = \pi i e^{2p\pi i},$$

令(**)式 $\delta \rightarrow 0, R \rightarrow \infty$ 可得 v.p. $\int_{-\infty}^{\infty} \frac{e^{px}}{1-e^x} dx = \pi i \frac{e^{2p\pi i} + 1}{e^{2p\pi i} - 1} = \pi \cot(p\pi),$

所以 v.p. $\int_{-\infty}^{\infty} \frac{e^{px} - e^{qx}}{1-e^x} dx = \text{v.p.} \int_{-\infty}^{\infty} \frac{e^{px}}{1-e^x} dx - \text{v.p.} \int_{-\infty}^{\infty} \frac{e^{qx}}{1-e^x} dx = \pi [\cot(p\pi) - \cot(q\pi)].$

当 x 充分大时, 有 $e^x - 1 \geq \frac{1}{2}e^x$ (即 $e^x \geq 2$)。不妨假设 $p > q$,

$$\left| \frac{e^{px} - e^{qx}}{1-e^x} \right| \leq 2 \frac{e^{px} - e^{qx}}{e^x} = 2[e^{-(1-p)x} - e^{-(1-q)x}]. \text{ 由于 } \int_0^{\infty} [e^{-(1-p)x} - e^{-(1-q)x}] dx \text{ 收敛, 所以}$$

$\int_0^{\infty} \frac{e^{px} - e^{qx}}{1-e^x} dx$ 收敛, 类似地, $\int_{-\infty}^0 \frac{e^{px} - e^{qx}}{1-e^x} dx$ 收敛, 所以原积分收敛, 等于其主值。